

MEMORANDUM

TO: Project Research Team (Domains 1, 2, and 3)

SUBJECT: Phase 1 Research Synthesis: Foundational Insights on Topology-Aware Road Extraction

PHASE: Transitioning from Phase 1 (Foundation) to Phase 2 (Divergent Deep Dive)

1. Executive Summary & Key Findings

Following our Phase 1 baseline review of the current literature, it is clear that while deep learning has revolutionized remote sensing, traditional pixel-level segmentation models (like standard U-Nets and DeepLab) frequently fail to capture the functional connectivity of road networks.

The primary finding for our project is that high pixel accuracy (IoU) does not guarantee a navigable map. Roads are often predicted with fragmented gaps due to occlusions (trees, shadows) or the inherent limitations of localized convolution operations. To achieve our project goal, we must shift focus from classifying isolated pixels to extracting mathematically sound, continuous vector graphs.

2. Best Models & Architectures (Domain 2 Focus)

To extract functional road topology, the state-of-the-art literature points to three primary architectural strategies that move beyond raw segmentation:

- **Graph Representation & GNNs:** These models treat the road network as an undirected graph $G(V,E)$, explicitly extracting intersections (nodes) and road segments (edges). Advanced networks utilize multi-task branches to simultaneously predict semantic segmentation, node heatmaps, and adjacency matrices to enforce connectivity.
- **Iterative Tracing Algorithms:** Instead of predicting the whole map at once, these models act like a digital pen. Starting from an initial vertex, the algorithm sequentially predicts the next point and direction. Recent state-of-the-art models (like RNGDet) fuse CNNs with Transformers to highly refine this step-by-step tracing.
- **Global Context Models (Transformers & Mamba):** Standard CNNs struggle with the narrow, elongated nature of roads stretching across a massive image. Vision Transformers (ViTs) and newer Mamba architectures (which use linear time complexity) are emerging as critical components. Their self-attention mechanisms capture long-range dependencies, allowing the model to understand that a road segment on the left side of the image connects to a segment on the right, bridging occluded gaps.

3. Identified Research Gaps

The foundational review identified four major bottlenecks in the current state-of-the-art that represent opportunities for our project to innovate:

1. **The Multi-Modal Gap:** Relying strictly on 2D optical imagery limits performance in complex urban environments. Fusing optical data with GPS trajectories or LiDAR is a highly promising area for improving robustness.
2. **The Annotation Bottleneck:** Creating perfect pixel-level ground truth data is prohibitively expensive. There is a major shift toward semi-supervised learning and weak labels, such as utilizing OpenStreetMap centerlines or the Segment Anything Model (SAM) to generate training topologies.
3. **Environmental Adaptability:** Models suffer from severe "domain shift" when trained in one geographic region and tested in another.
4. **Computational Efficiency:** Topological and graph-based networks are incredibly memory-intensive; there is a critical need for lightweight network designs.

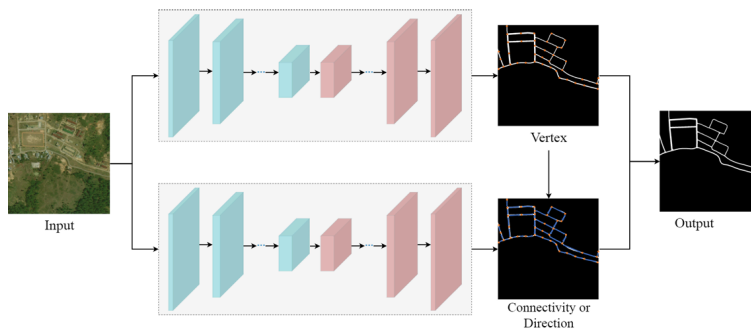


Figure 8. The general architecture of the Graph model.

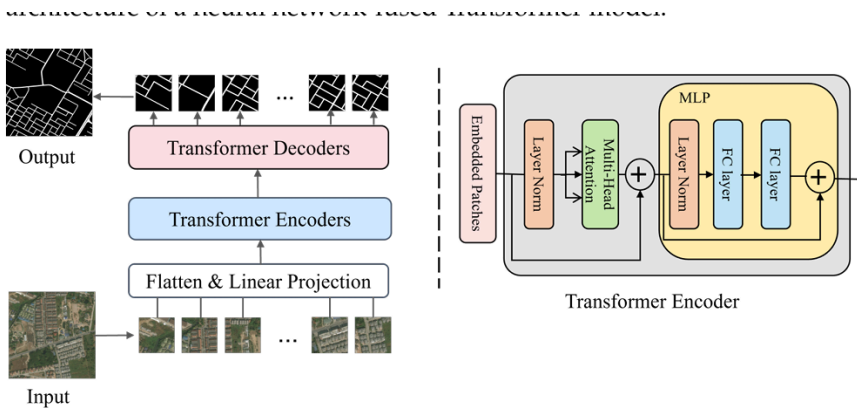


Figure 9. The general architecture of a neural network-fused Transformer model.